

IE 214 LP Formulation

The location of the homebase station of an Autonomous Mobile Robot (AMR) needs to be determined in the Intelligent Manufacturing Lab at the IE/ME Building. The AMR which only moves in a horizontal or vertical direction, will service four different workstations in the Lab. The facilities planning engineer has been designated to accomplish this task and the engineer must specify the coordinates (x_1, x_2) of the optimal location of the homebase station.



The four workstations at the Lab are located at the following coordinates:

Workstation Alpha	(1,10)	Workstation Charlie	(4,1)
Workstation Bravo	(6,5)	Workstation Delta	(-6,-5)

Formulate the problem of finding an optimal location of the homebase station as an LP if the sum of the distances it can travel from the homebase station to the existing workstations is to be minimized. Since the AMR can only travel in a horizontal or vertical direction, the distance traveled from an arbitrary point (x_1, x_2) to Workstation Alpha at (1,10) is $|x_1 - 1| + |x_2 - 10|$. The distance to the other workstations can be calculated in a similar fashion. This means that the distance is not defined by the length of the line segment between the two points, rather it is the sum of the lengths of the horizontal and vertical components of such a line segment.